incorrect wiring of speakers is a thing of the past.

Let The Tests Begin!

For the Intel platform, the motherboards fell within the Rs 5,000 to Rs 10,000 range. Asus, MSI, Gigabyte and AOpen managed to send us boards, most of them based on the 915G chipset. We did receive boards based on the RS350 and 865 chipsets, but we deferred to test them since they are based on the older socket 478 processors and would be obsolete soon. Of the seven boards based on 915G, MSI sent us three, Gigabyte two, while Asus managed to send one. Unfortunately, the AOpen was ruled out due to a malfunction.

Features

All these motherboards support the latest Pentium 4 800 MHz LGA 775 processors, clocked up to 3.8 GHz. Care must be taken while installing the processor, as the pins are on the motherboard socket. The heatsink/fan unit does not have a plastic retention mechanism like the one on older Socket 478 boards. So check the heatsink for proper contact with the processor after installation.

Gigabyte's GA-8I915G-MF and MSI 915G Neo 2 score high for their clean and unhindered layout—no capacitors near the processor area. MSI provides a unique CPU installation clip so as to avoid damage to the motherboard socket.

MSI's 915G combo and Gigabyte's GA-8I915G Duo support the DDR and DDR II standards. Both these boards have four slots—two each for DDR and DDR II—to ensure dual-channel memory support. The Platinum series boards from MSI only support the DDR II standard, whereas the Asus P5GD1-VM and Gigabyte GA-8I915G-MF support DDR 400 MHz as well.

Since most 915 boards use the ICH6 Southbridge, almost all boards support SATA drives. However, MSI's Platinum boards use the ICH6R Southbridge, thus adding RAID 0, 1, 0+1 and Intel's Matrix RAID capability to the storage sub-system. If you plan to use the system for I/O intensive work, requiring faster data transfers and reliability, you might want to look at these boards; otherwise the above are just extra features that you won't use.

While SATA is increasing in presence, support for older IDE connectors is still in place. The Southbridge still supports one IDE drive, and hence, connecting your old CD-ROM is not a problem—yet. The ASUS P5GD1-VM and Gigabyte GA-8I915G-MF have just one IDE connector, whereas MSI's Platinum boards and Gigabyte's GA-8I915G Duo have three—one via the Southbridge, and the other two, RAID-capable, via a third-party controller, namely the VIA VT-6410.

Since MSI's 915P Neo 2 Platinum is based on the 'P' version of the 915 chipset, it lacks an onboard video controller, while the other boards have an integrated GMA 900 video chip. Every board in this mini round-up features the x16 PCI-Express lane for plugging a discrete video card, if you need to play high-end games. Except for the Asus P5GD1-VM with just one x1 PCI-Express slot, all the other boards featured a minimum of two x1 lanes.

On the peripherals side, all motherboards now come with eight USB2.0 ports and Gigabit network adapters for faster data transfer. FireWire ports still evade the motherboards in general, with only the Platinum series motherboards from MSI featuring them.

As for the sound system, all boards had onboard support for Intel's High Definition 7.1-channel audio. Almost all boards feature PCI slots for those older sound cards and network cards.

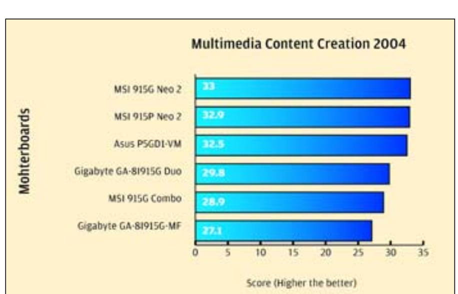
What we can conclude from this is that no motherboard comes close to the MSI 915G and 915P Neo 2 Platinum in terms of features offered. Gigabyte's GA-8I915G Duo and MSI's 915G Combo combine assorted features so as to keep the cost down. Asus' P5GD1-VM and the Gigabyte's GA-8I915G-MF have all the essential features.

Performance

Let's take one benchmark at a time to see how each motherboard fared.

Content Creation 2004

MSI's 915G Neo 2 does a great job, returning a score of 33. Close on its heels is its sibling, the 915P Neo 2. Third place is taken by Asus' P5GD1-VM. Gigabyte's GA-8I915G0MF scored 17 per cent less than MSI's top board. All this shows that MSI's 915G Neo 2 has a well-oiled sub-system.

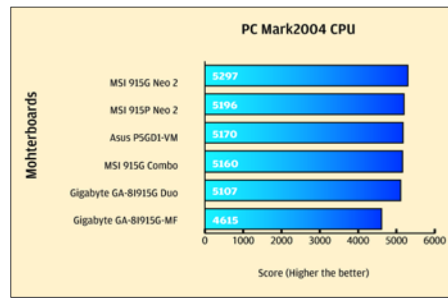


PCMark 2004

This test suite is also a system-level benchmark, testing both the processor and I/O performance.

The CPU Sub-system

MSI's 915G Neo 2 was way ahead of the other boards when it came to the CPU sub-system, and the CPU sub-system was the reason for this board's good performance in the Content Creation test.



Overall, the results are pretty similar with second place once again going to the MSI 915P Neo 2, third place being secured by Asus' P5GD1-VM. Both the Gigabyte boards are in the last spot. A point to observe is that the performance of the GA-8I915G-MF board is nearly 12 per cent poorer than MSI's 915G Neo 2 in the CPU sub-system test, resulting in the low Content Creation score.

The I/O Sub-system

In the I/O sub-system, Gigabyte's GA-8I915G Duo gains over MSI's Platinum boards by a small margin. MSI's 915G Combo, without a doubt, has a poor I/O subsystem, a fact reflected in its Content Creation score. Asus' P5GD1-VM also scores less on the I/O. These are not the boards to look at if your work involves image or video manipulation.

Memory Test

Gigabyte's GA-8I915G Duo again scored over the MSI Platinum motherboards—by a small margin of 0.22 per cent. Asus' P5GD1-VM and MSI 915G Combos were right at the bottom of the score sheet, suggesting a CPU-memory link inferior to the Gigabyte board.